culturecollection of algae and protozoa

media recipes

MErds / MY75S

Medium

Biphasic. See separate recipes.

MErds (Modified Føyns Erdschreiber Medium)

Marine protozoa

Stock		**per 100 ml**
(1)	$NaNO_3$	20.0 g
	Na_2HPO_4	1.2 g

Medium	**per litre**
Soil extract with salts (SES) - see below	100.00 ml
Stock solutions (1) and (2)	1.0 ml each
Filtered seawater	898.0 ml

Mix the above constituents and autoclave at 15 psi for 15 minutes. It may be necessary to filter final medium to avoid problems with precipitate.

SES (Soil Extract with Added Salts)

Stocks		**per litre**
(1)	K_2HPO_4	1.0 g
(2)	$MgSO_4.7H_2O$	1.0 g
(3)	KNO_3	10.0 g

Medium	**per litre**
Stock solutions 1 - 3	20.0 ml each
Soil extract (*SE - see overleaf)	100.0 ml

* At the CCAP, SE1 is used for marine algae, SE2 for freshwater and terrestrial protozoa.

Make up to 1 litre with deionized water and autoclave at 15 psi for 15 minutes.

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SE1 (Soil Extract 1)

used in media for marine algae and protozoa

Preparing the soil

Site selection for a good soil is very important and for most purposes a soil from undisturbed deciduous woodland is best. Sites to avoid are those showing obvious signs of man's activity and particular care should be taken to avoid areas where fertilizers, crop sprays or other toxic chemicals may have been used.

A rich loam with good crumb structure should be sought. Stones, roots and larger invertebrates should be removed during an initial sieving through a 1 cm mesh. The sieved soil should be spread to air dry and hand picked for smaller invertebrates and roots. It should be turned periodically and picked over again. When dry it may be sieved through a finer mesh (2-4 mm) or stored as it is prior to use.

Medium

Soil is prepared as above. Air-dried soil and twice its volume of supernatant distilled water are autoclaved together at 15 psi for 2 hours and left to cool. The supernatant is then decanted and filtered through Whatman No 1 filter paper, then distributed to containers in volumes suitable for making up batches of media. The aliquots and their containers are autoclaved for an appropriate length of time (e.g. 1 litre or less for 15 minutes) and are then kept in a cool place (e.g. a refrigerator) until required.

MY75S (Malt & Yeast Extract - 75% Seawater Agar)

Marine Protozoa

Medium	**per litre**
Natural seawater, filtered GF/C	750.0 ml
Deionized water	250.0 ml
Malt extract (Oxoid L39) *	0.1 g
Yeast extract (Oxoid L21) *	0.1 g
Bacteriological Agar (Oxoid L11) *	15.0 g

Thoroughly disperse the agar in cold 75% seawater. Bring to the boil, stirring continuously. During heating add the other constituents. Transfer the molten agar to suitable vessels and sterilise at 15 psi for 15 minutes. If available a bench top agar maker is preferable to making this medium. The constant stirring during sterilisation minimises flocculation.

Supply

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